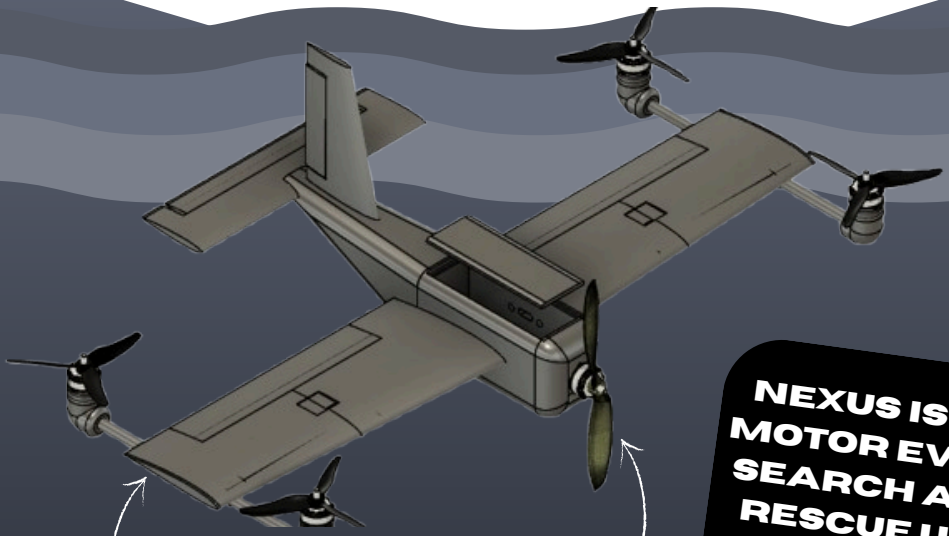
S.AND.R NEXUS INSTRUCTION MANUAL

AIDEN ERVIN



NEXUS IS A 5 MOTOR EVTOL SEARCH AND RESCUE UAV PLATFORM

NEXUS IS CAPABLE OF COMPLETING SEMI AUTONOMOUS WAYPOINT MISSIONS

NEXUS IS EQUIPPED WITH A CUSTOM COMPUTER VISION MODEL FOR SEARCH AND RESCUE MISSIONS

REPO

PROJECT BY - AIDEN ERVIN
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17
TN USA

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GENERAL AIRCRAFT DATA

DIMENSIONS TABLE

WINGSPAN	24 INCHES
HALFSPAN	12 INCHES
ROOT CHORD	5 INCHES
TIP CHORD	4.25 INCHES
WING AREA	111 IN SQUARED
MTOW	1.7LBS
WING LOAD	0.23 OZ/IN SQUARED

BILL OF MATERIALS

BILL OF MATERIALS

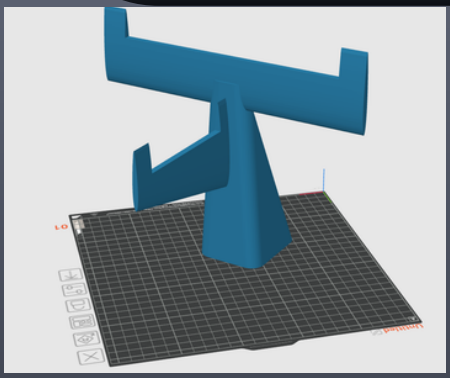
BOM LINK

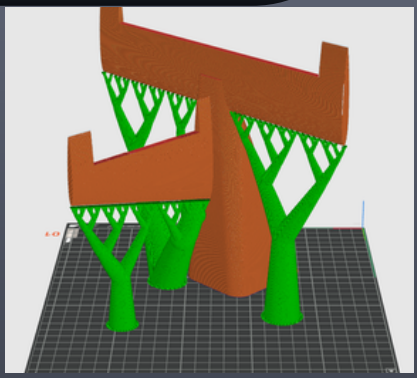
Part	Quantity	Cost
Flight Controller	1	$75.53
Pusher Prop	1 Prop	$7.09
VTOL Props	1 (Pack) 4 Props Total	$13.19
XT30 Pigtails (M)	5	$6.74
ELRS Reciever	1	$8.29
XT30 PDB Plugs (F)	5	$6.23
PDB Board	1	$6.21
XT30 Pigtail Pair (M&F)	1 Pair	$0.99
VTOL ESC	1 Pack 4 ESC's	$27.30
Pusher ESC	1 ESC	$1.20
Pusher Motor (CCW)	1	$4.53
VTOL Motors	4	$23
Control Surface Servos	4	$21.71
GPS	1	$11.96
Battery	1	$17.99
Filament (PETG)	1	$13.99
Filament (LW-PLA)	1	$29.99
Camera & VTX Combo	1	$45.99
CA Glue And Activator	1	$21.29
Solder Flux	1	$8.49
Heat Inserts And Screws	1	$2.17
Heat Shrink	1	$7.99
Servo Linkages	1Pack	$0.99
Control Horns	1 Pack	$0.99
Control Surface Hinges	1 Pack	$10
Carbon Fiber Booms 10mm OD	1 Tube Will be Cut Into 2	$23.80
Carbon Fiber Booms 6mm OD	2 Tubes Total	$31
Total		**$428.65**

PRINTED PARTS

PRINTED PARTS LIST

TAIL ORIENTATION GYROID INFILL PATTERN





Layer height	Wall loops	Infill density(%)	Support	Brim	Outer wall speed
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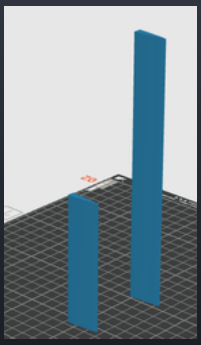
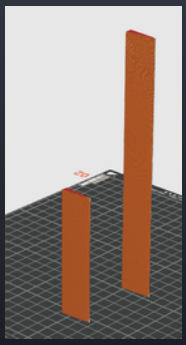
TAIL CONTROL SURFACES ORIENTATION





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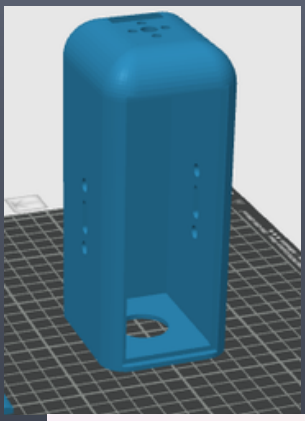
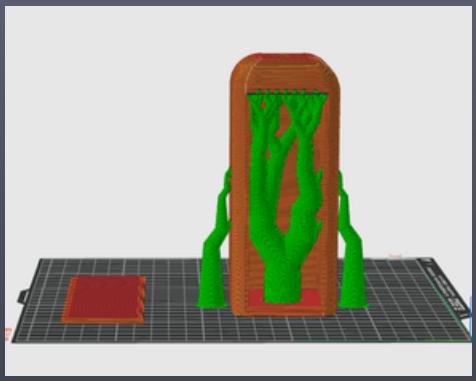
GYROID INFILL

PRINT ORIENTATION AND PARAMETRS

FUSELAGE POD AND HATCH ORIENTATION

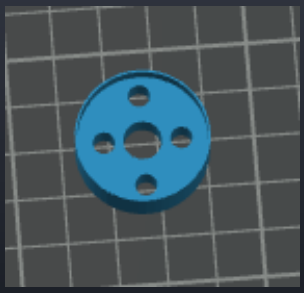




GYROID INFILL

Layer height	Wall loops	Infill density(%)	Support	Brim	Outer wall speed
0.20	3	10.0	☑	No-brim	160

VTOL MOTOR MOUNT CAP ORIENTATION



Layer height	Wall loops	Infill density(%)	Support	Brim	Outer wall speed
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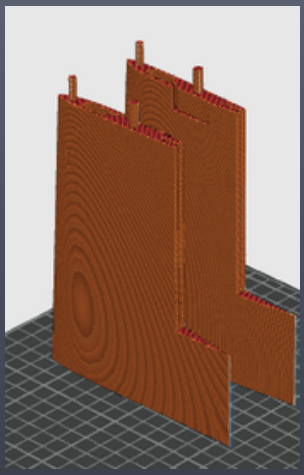
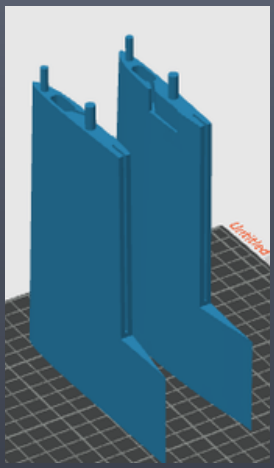
PRINT ORIENTATION AND PARAMETRS

WING 1 AND 2 ORIENTATION





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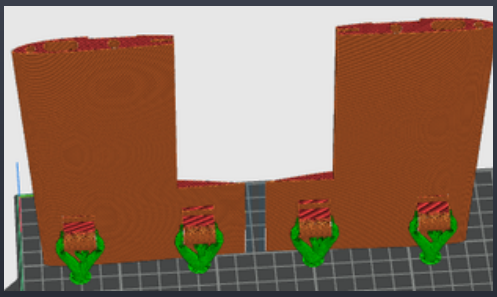

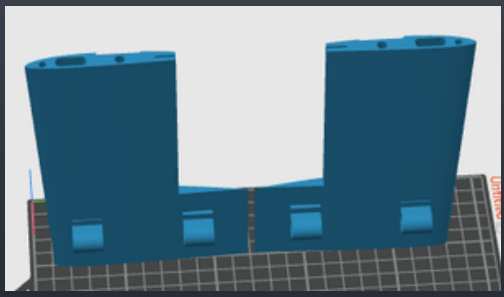

Layer height	Wall loops	Infill density(%)	Support	Brim	Outer wall speed
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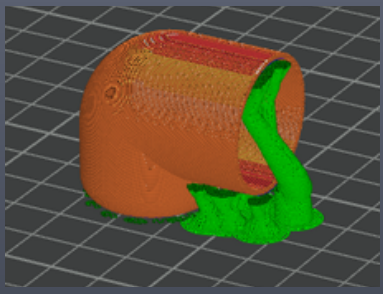
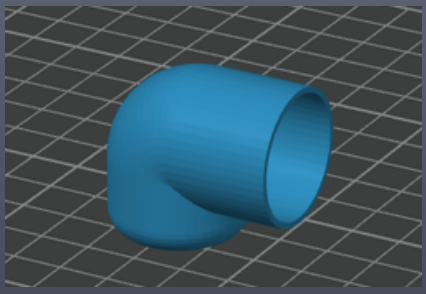
PRINT ORIENTATION AND PARAMETRS

VTOL MOTOR MOUNT ORIENTATION





Layer height	Wall loops	Infill density(%)	Support	Brim	Outer wall speed
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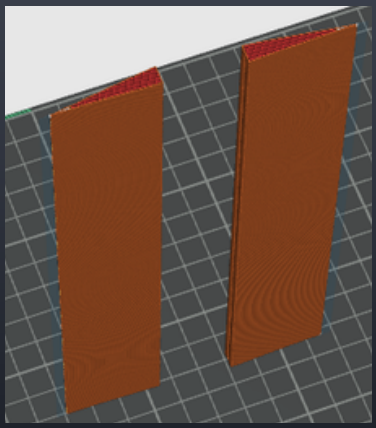
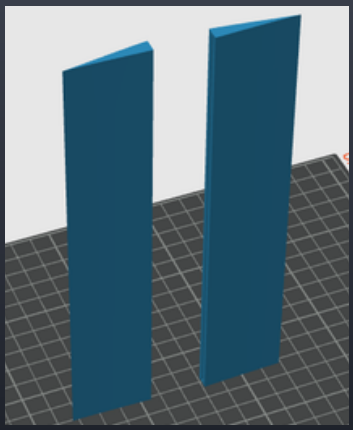
AILERON ORIENTATION



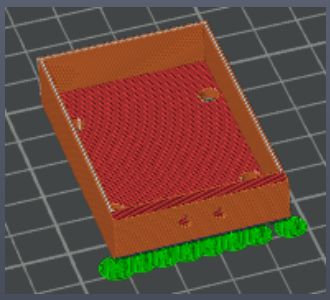


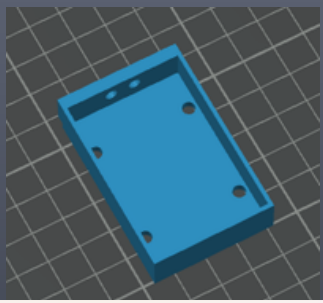
Layer height	Wall loops	Infill density(%)	Support	Brim	Outer wall speed
0.20	3	10.0	☑	No-brim	160

GYROID INFILL

PRINT ORIENTATION AND PARAMETRS

FLIGHT CONTROLLER MOUNT ORIENTATION

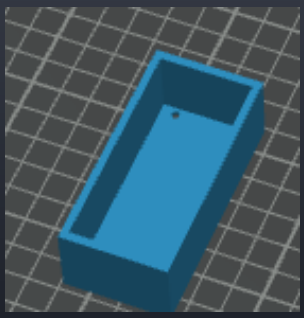


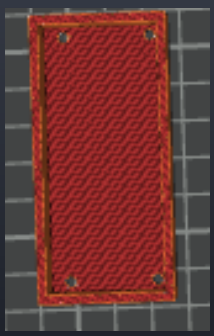


GYROID INFILL

Layer height	Wall loops	Infill density(%)	Support	Brim	Outer wall speed
0.20	3	10.0	☑	No-brim	160

BATTERY PAD ORIENTATION





GYROID INFILL

Layer height	Wall loops	Infill density(%)	Support	Brim	Outer wall speed
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FUSELAGE AND TAIL ASSEMBLY

CONNECTING THE 2 FUSELAGE AND TAIL PIECES

ELAVATOR AND RUDDER CONNECTION

SERVO SETUP AND WIRING

FLIGHT CONTROLLER MOUNT SETUP

BATTERY TRAY SETUP

WING ASSEMBLY

CONNECTING THE 2 WING SECTIONS

AILERON CONNECTION

SERVO SETUP AND WIRING

VTOL SETUP

VTOL SETUP

AUTOPILOT SETUP

DOWNLOADING ARDUPILOT AND FLASHING ARDUPLANE

PARAMATER SETUP

RECIEVER SETUP

ELECTRONICS SETUP

FLIGHT CONTROLLER PLACEMENT

GPS, RECIEVER AND TELEMETRY SETUP

MOTOR AND ESC SOLDERING

FIRST PLUG IN

MAIDEN FLIGHT

MAIDEN FLIGHT CHECK LIST